

Experiment/ Case Study No.:3

1. **Experiment Title:** Implement Control Structures and Form Validation; Create a Grading System Based on User-Entered Marks

2. **Software/Tools Required:** VS Code, Web Browser, HTML, CSS, JavaScript

3. **Experiment Program Code:**

grading.css

```
body {
    font-family: Arial, sans-serif;
    margin: 20px;
    text-align: center;
    background-color: #f8f9fa;
    color: #333333;
}
h2 {
    color: #28a745;
}
form {
    margin: 0 auto;
    width: 300px;
    text-align: left;
    background-color: #ffffff;
    padding: 20px;
    border: 1px solid #dee2e6;
    border-radius: 4px;
}
input[type="text"], input[type="number"], input[type="button"] {
    margin-top: 5px;
    margin-bottom: 15px;
    display: block;
    width: 100%;
    box-sizing: border-box;
    padding: 8px;
    border: 1px solid #ced4da;
    border-radius: 4px;
}
input[type="button"] {
    background-color: #28a745;
    color: white;
    border: none;
    cursor: pointer;
    font-weight: bold;
}
input[type="button"]:hover {
    background-color: #218838;
}
.report-card {
    display: none;
    margin: 20px auto 0 auto;
    text-align: left;
    width: 300px;
    background-color: #d4edda;
    border: 1px solid #c3e6cb;
    color: #155724;
    padding: 20px;
    border-radius: 4px;
}
```

grading.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Student Grading System</title>
    <link rel="stylesheet" href="grading.css">
</head>
<body>
    <h2>Student Grading System</h2>
    <form>
        <label>Student Name:</label>
        <input type="text" id="name">
        <label>Marks: </label>
        <input type="number" id="marks">
        <input type="button" value="Calculate Grade" onclick="GradeSystem()">
    </form>
    <div id="report" class="report-card"></div>
    <script src="grading.js"></script>
</body>
</html>
```

grading.js

```

function GradeSystem() {
    var name = document.getElementById("name").value;
    var marks = document.getElementById("marks").value;

    if (name == "") {
        alert("Please enter Student Name");
        return;
    }

    if (marks == "") {
        alert("Please enter marks");
        return;
    }

    marks = Number(marks);
    if (marks < 0 || marks > 100) {
        alert("Invalid marks");
        return;
    }

    var grade;
    var result;
    if (marks >= 90) { grade = "A+"; result = "Pass"; }
    else if (marks >= 80) { grade = "A"; result = "PASS"; }
    else if (marks >= 70) { grade = "B"; result = "PASS"; }
    else if (marks >= 60) { grade = "C"; result = " Pass"; }
    else if (marks >= 50) { grade = "D"; result = "Pass"; }
    else if (marks >= 40) { grade = "E"; result = "Fail"; }
    else { grade = "F"; result = "Fail"; }

    var reportDiv = document.getElementById("report");
    reportDiv.innerHTML = "<h2>Student Grade Report</h2>" +
        "<br>" +
        "<b>Student Name :</b> " + name + "<br><br>" +
        "<b>Marks :</b> " + marks + "<br><br>" +
        "<b>Grade :</b> " + grade + "<br><br>" +
        "<b>Result :</b> " + result;
    reportDiv.style.display = "block";
}

```

4. Output:

5. Case Study Title: Password Validation Webpage using Form Validation and Control Structures

6. Case Study Program Code:

login.css

```
body {
  font-family: Arial, sans-serif;
  margin: 20px;
  text-align: center;
  background-color: #f8f9fa;
  color: #333333;
}
h2 {
  color: #28a745;
}
form {
  display: inline-block;
  text-align: left;
  background-color: #ffffff;
  padding: 20px;
  border: 1px solid #dee2e6;
  width: 300px;
  border-radius: 4px;
}
input[type="text"], input[type="password"] {
  display: block;
  margin-top: 5px;
  margin-bottom: 10px;
  width: 100%;
  box-sizing: border-box;
  padding: 8px;
  border: 1px solid #ced4da;
  border-radius: 4px;
}
input[type="submit"] {
  background-color: #28a745;
  color: white;
  border: none;
  padding: 10px;
  cursor: pointer;
  width: 100%;
  margin-top: 10px;
  font-weight: bold;
  border-radius: 4px;
}
input[type="submit"]:hover {
  background-color: #218838;
}
.error {
  color: #dc3545;
  font-size: 13px;
  margin-bottom: 10px;
}
.success {
  color: #28a745;
  font-weight: bold;
  margin-top: 10px;
  text-align: center;
}
```

login.js

```

function validateForm() {
    var loginId = document.getElementById('loginId').value.trim();
    var password = document.getElementById('password').value;

    var loginIdError = document.getElementById('loginIdError');
    var passwordError = document.getElementById('passwordError');
    var successMessage = document.getElementById('successMessage');

    // Reset
    loginIdError.innerHTML = '';
    passwordError.innerHTML = '';
    successMessage.innerHTML = '';

    var isValid = true;

    // Validate Login ID
    if (loginId === '') {
        loginIdError.innerHTML = 'Login ID is required.';
        isValid = false;
    } else if (loginId.length < 3) {
        loginIdError.innerHTML = 'Login ID must be at least 3 characters.';
        isValid = false;
    } else if (!/^[a-zA-Z\s]+$/i.test(loginId)) {
        loginIdError.innerHTML = 'Login ID can only contain letters and spaces.';
        isValid = false;
    }

    // Validate Password
    if (password === '') {
        passwordError.innerHTML = 'Password is required.';
        isValid = false;
    } else {
        var pwdErrors = [];
        if (password.length < 8) pwdErrors.push('at least 8 characters');
        if (!/[A-Z]/.test(password)) pwdErrors.push('one uppercase letter');
        if (!/[a-z]/.test(password)) pwdErrors.push('one lowercase letter');
        if (!/[0-9]/.test(password)) pwdErrors.push('one integer');
        if (!/[!@#%&*(),.?":{}|<>]/.test(password)) pwdErrors.push('one special character');

        if (pwdErrors.length > 0) {
            passwordError.innerHTML = 'Password must contain: ' + pwdErrors.join(', ') + '.';
            isValid = false;
        }
    }

    if (isValid) {
        successMessage.innerHTML = 'Login Successful!';
    }

    return false;
}

```

student-login.html

```

<!DOCTYPE html>
<html>
<head>
    <title>SIT Student Login</title>
    <link rel="stylesheet" href="login.css">
</head>
<body>
    <h2>SIT Student Login</h2>
    <form onsubmit="return validateForm()">
        <label>Login ID (Name):</label><br>
        <input type="text" id="loginId"><br>
        <div class="error" id="loginIdError"></div>

        <label>Password:</label><br>
        <input type="password" id="password"><br>
        <div class="error" id="passwordError"></div>

        <input type="submit" value="Login">
        <div class="success" id="successMessage"></div>
    </form>
    <script src="login.js"></script>
</body>
</html>

```

8. Output:

9. Result/Conclusion:

The practical was completed successfully. Control structures (if-else, if-else-if, for loops, ternary operator) and form validation were implemented using JavaScript. A password validation system (Task 3.b) and a grading system (Task 3.a) were developed to validate user input and display the appropriate grade based on entered marks.